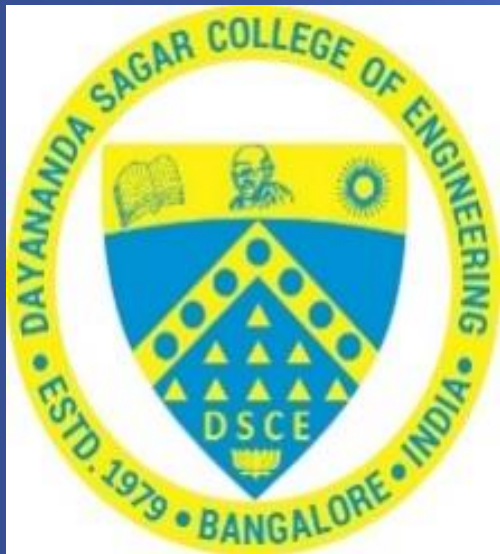


Fundamentals of Robotics and Applications

Unit 1: Introduction



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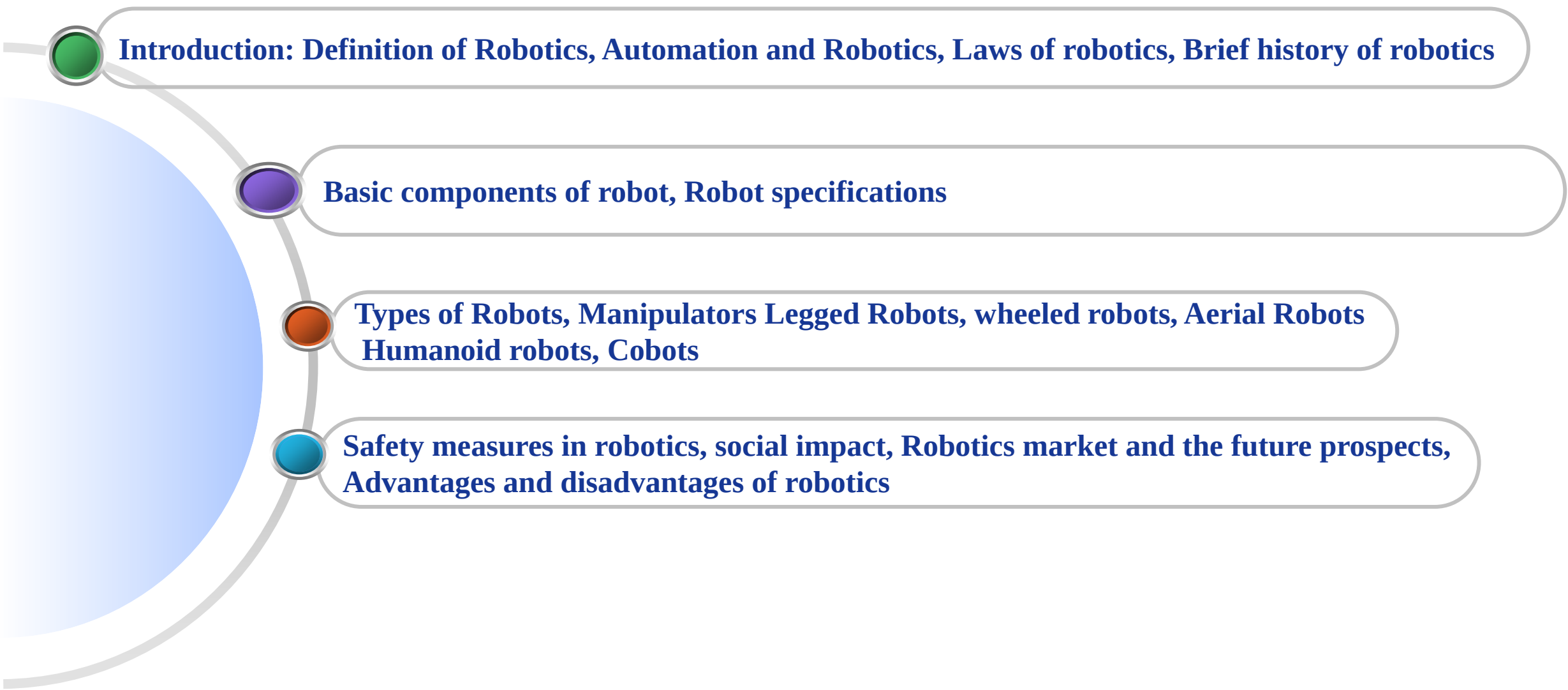
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Introduction: Definition of Robotics, Automation and Robotics, Laws of robotics, Brief history of robotics

Basic components of robot, Robot specifications

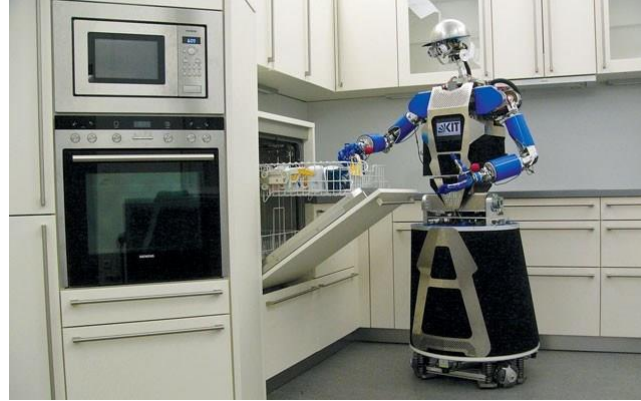
**Types of Robots, Manipulators Legged Robots, wheeled robots, Aerial Robots
Humanoid robots, Cobots**

**Safety measures in robotics, social impact, Robotics market and the future prospects,
Advantages and disadvantages of robotics**

Introduction

What comes to your mind when you first hear or think of a robot ?

- Mechanical machines ?
- Companions / friends
- Pets / toys
- Helpers
- Drones/UAV



PAST- Industrial Robot Arms



- 1960 Dangerous work used to do by robotics
- Example spot welding

PRESENT- Life of an amazon robot

- Lakhs of parts picked by 3000 to 4000 mobile robots
- Movement should be optimal
- No collision
- Motion planning complicated



FUTURE- Social robot

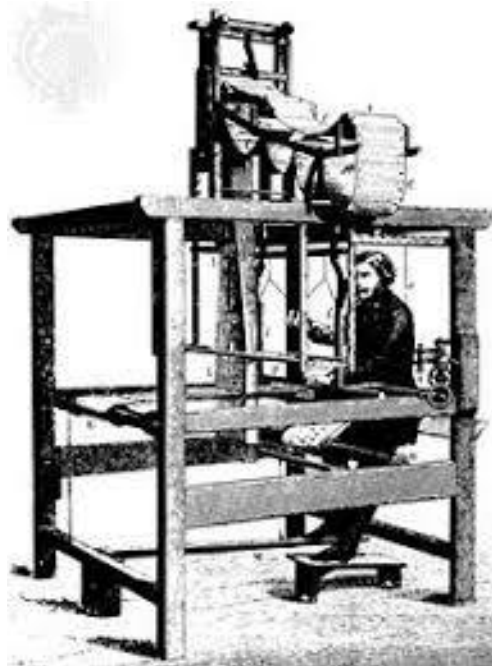
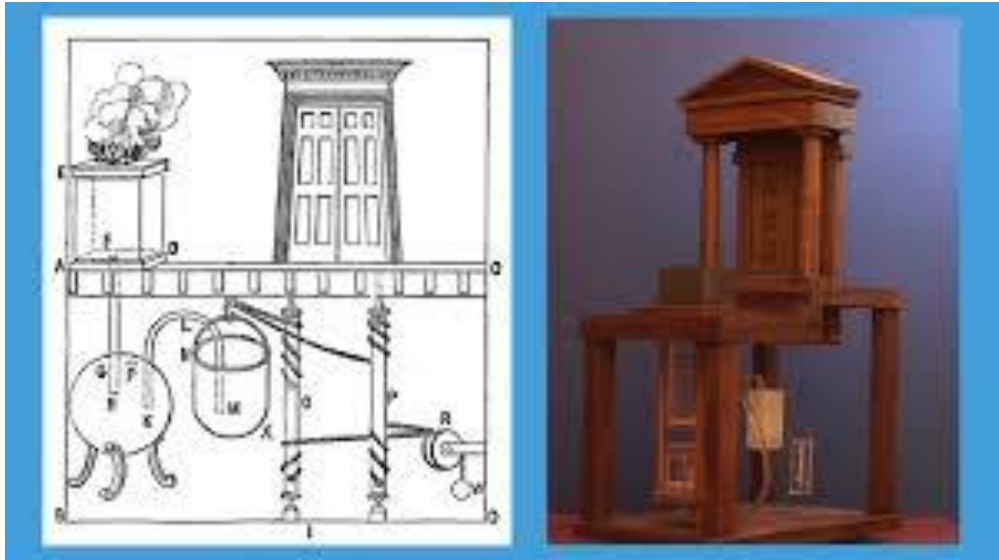


- Society- railway station, schools, hospitals, restrauunts
- Clones- Robots That Look like Humans
- Challenges- Safety, Intelligence (AI) Human like behavior and looks



Origin of Automation: Replacing human Muscle power

- 10,000 BC Stone tools used in early civilization.
- Design of simple automation (150 BC) moving engine, Herons door etc. in Greece.
- 1780 AD saw the creation of automatic dolls which could write, draw pictures etc.
- Punch cards used in power looms in France in 1801 for manufacture of textiles Joseph-Marie-Jacquard.



Hard Automation in Ford Motor Company 1904

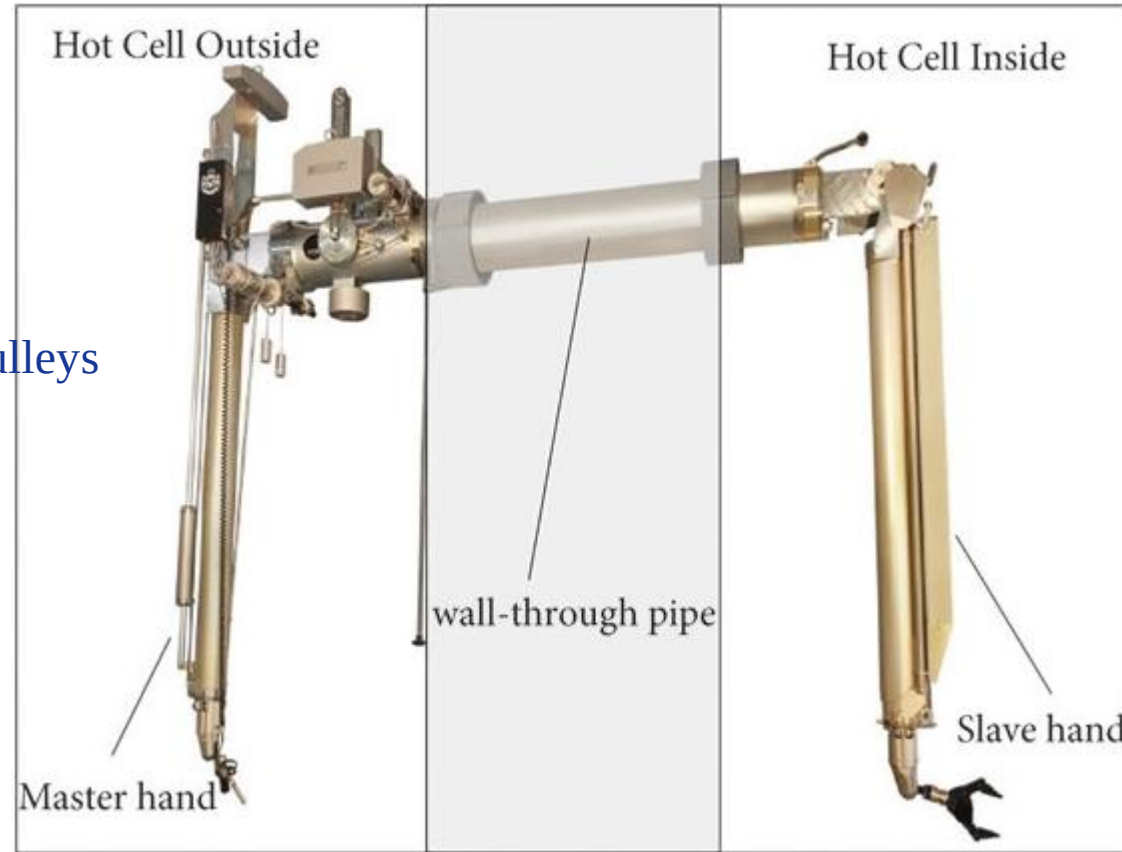
- Idea of transfer lines in which a car was assembled at different stations.
- First use of hard automation – alignment devices, transfer devices etc
- 1904 Henry Ford's mass production of vehicles in the USA



History

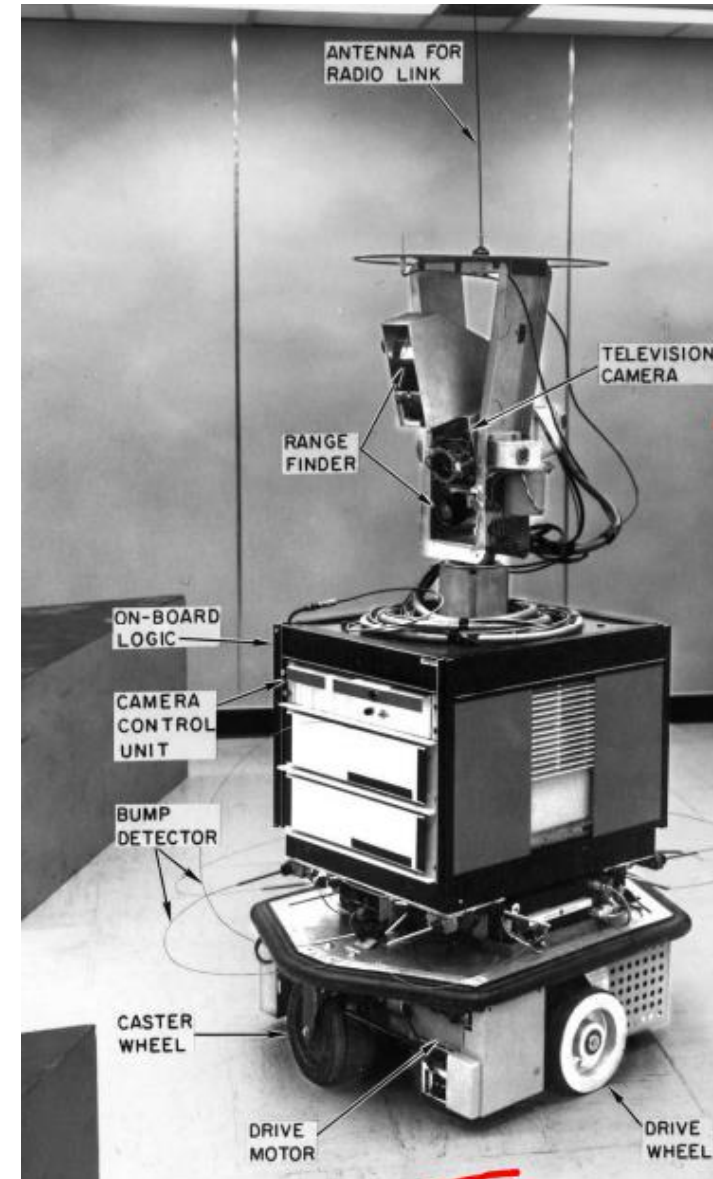
- 1921 Karel Kapec's play depicting human like mechanical man (slave) – robots
- 1942 Isaac Asimov first used the term Robotics
- Origin of robots: 1945 master slave manipulator made for radioactive material handling for the Atom Bomb project.

A strictly mechanical device
Motion transfer by wire rope and pulleys

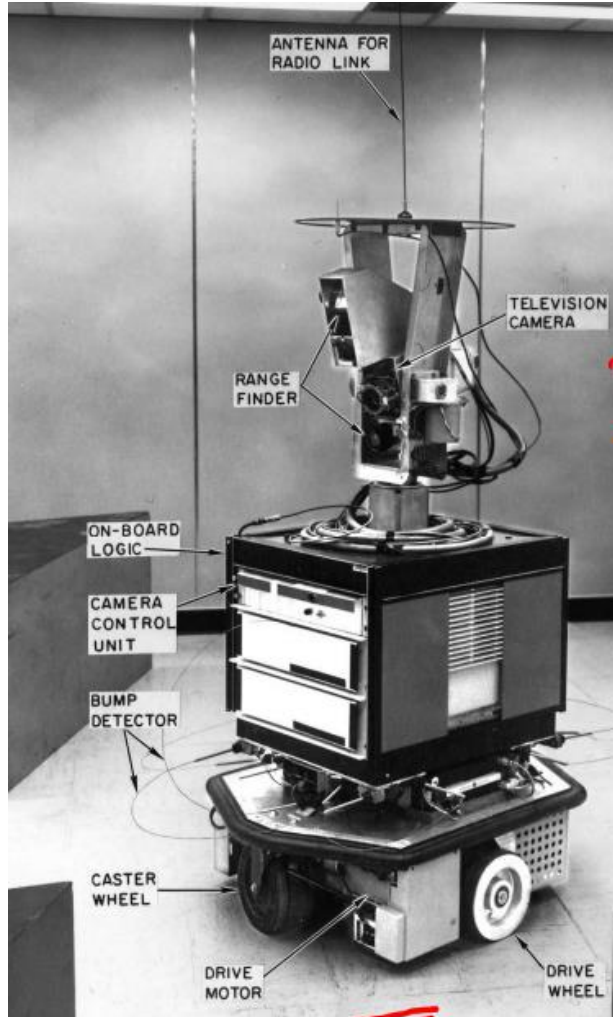


What changed everything ?

- Mechanical systems became electro-mechanical !
- Transistor - (1949) : concept of reprogram!
- 1950 SHAKY : First robot (MOBILE ROBOT-AVOID OBSTACLE)
- 1952 George Dovel : teach / play back devices for NC machines/ robots
- Transistor-Computer-Micro
Computer-Microcontrollers-DSP



Clumsy robots to sophisticated humanoids



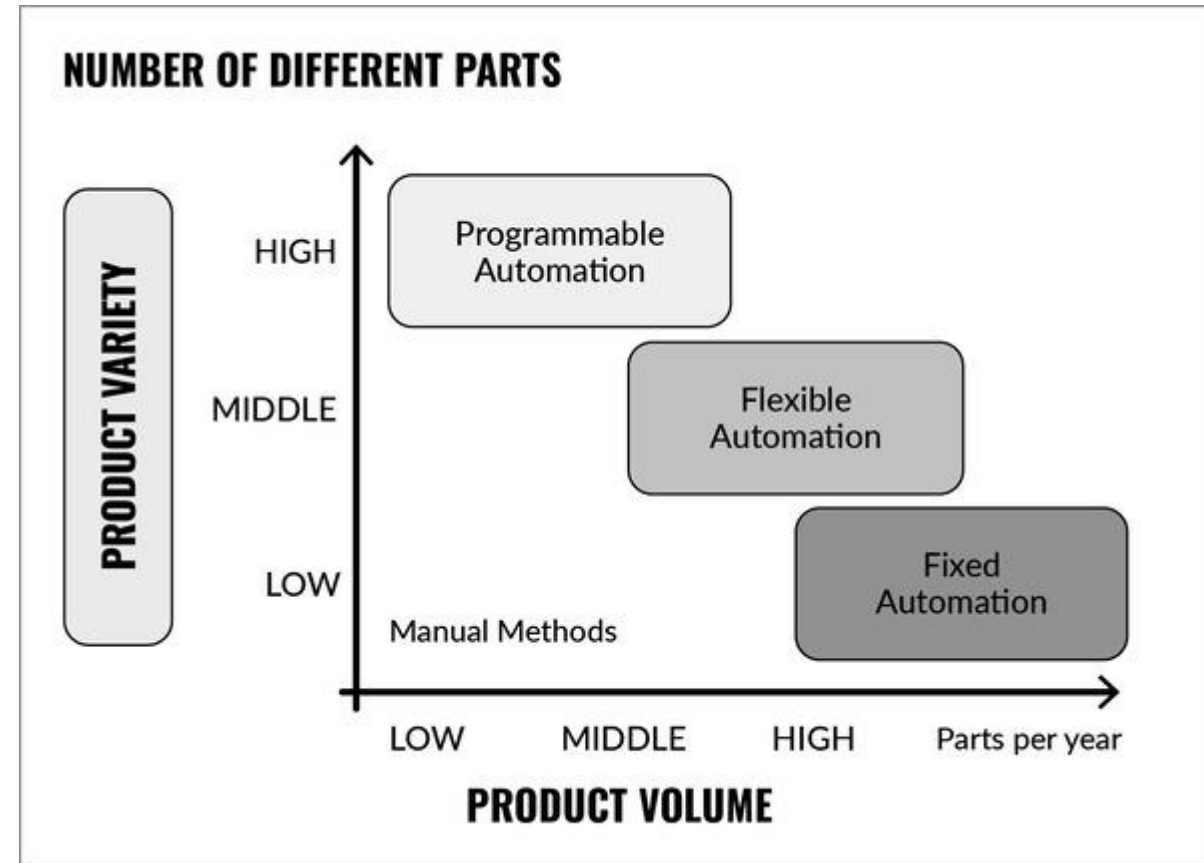
1950



2024 (54DOF)

Automation

- **Automation** : In industrial context, automation is defined as a technology that is concerned with the use of mechanical, electronic and computer based systems in the operation and control of production.
- Example: Transfer lines, mechanized assembly machines, feedback control systems, NC machine tools and robots.
- Robotics is a form of industrial automation.
- **Types of Automation:** Fixed automation, Programmable Automation, Flexible Automation.



Types of Automation

- **Fixed automation** is used when the volume of production is very high, high production rates, initial investment is high, specially designed to one type of product. Ex: Automobile industry, food packing industry
- **Programmable Automation** is used when the volume of production is relatively low and there are a variety of products to be made. Program is read into the production equipment and the equipment performs the particular sequence of processing(assembly) operations to make that product. Example CNC Machines, Industrial Robot.
- **Flexible Automation** or Flexible manufacturing system or CIM systems is suitable for mid-volume production range. An extension of programmable automation in which the system is capable of changing over from one job to the next with no lost time between jobs. The best examples FMS, CIM, Surgical Robots etc.

Industrial Robot and Robotics

- The official definition of an Industrial Robot is provided by Robotics Industries Association(RIA), formally the Robotics Institute of America
- “An Industrial Robot is a reprogrammable, multifunctional, manipulator designed to move materials, parts, tools, or special devices through variable programmed motions for the performance of a variety of tasks” . Therefore industrial robots should be classified as a form of programmable automation.
- Robots themselves are examples of Programmable automation, they are sometimes used in flexible automation and even fixed automation system.
- Robotics can be defined as a field of technology that deals with the conception, design, construction, operation and applications of Robots.

What is the definition of a Robot ?

- Today with the appearance of so many different types of robots ranging from mobile robots, bipeds, humanoids, cyborgs, UAVs, old definition of robot is no longer able to accommodate all of them.
- **To be called a robot it should do some or all of the following :**
- **- move around - sense and manipulate the environment and display intelligent behavior.**
- In recent trend robots are classified into industrial robots and service robots. An industrial robot is essentially a manipulator arm for automation, while service robots are those that operate autonomously to perform service useful to the well being of humans and equipment.

Differences between Robotics and Automation ?

- Robotics focuses on systems incorporating sensors and actuators that operate autonomously or semi-autonomously in cooperation with human
- Robotics research emphasizes intelligence and adaptability to cope with unstructured environments.
- Automation research emphasizes efficiency, productivity, quality, and reliability, focusing on systems that operate autonomously, often in structured environments over extended periods, and on the explicit structuring of such environments.

Laws of robotics

Three laws of robotics by Isaac Asimov are

1. A robot may not injure a human being or through inaction, allow a human to be harmed.
2. A robot must obey orders given by humans except when that conflicts with the First law.
3. A robot must protect its own existence unless that conflicts with the first or second laws.

Brief history of robotics

Date	Development
mid-1700s	J. de Vaucanson built several human-sized mechanical dolls that played music.
1801	J. Jacquard invented the Jacquard loom, a programmable machine for weaving threads or yarn into cloth.
1805	H. Maillardet constructed a mechanical doll capable of drawing pictures.
1946	American inventor G. C. Devol developed a controller device that could record electrical signals magnetically and play them back to operate a mechanical machine U.S. patent issued in 1952.
1951	Development work on teleoperators (remote-control manipulators) for handling radioactive materials. Related U.S. patents issued to Goertz (1954) and Bergsland (1958).
1952	Prototype Numerical Control machine demonstrated at the Massachusetts Institute of Technology after several years of development. Pan programming language called APT (Automatically Programmed Tooling) subsequently developed and released in 1961.
1954	British inventor C. W. Kenward applied for patent for robot design. British patent issued in 1957.
1954	G. C. Devol develops designs for "programmed article transfer." U.S. patent issued for design in 1961.
1959	First commercial robot introduced by Planet Corporation. It was controlled by limit switches and cams.

Contd.

Brief history of robotics

10

Industrial Robotics

1960	First "Unimate" robot introduced, based on Devol's "programmed ankle transfer." It used numerical control principles for manipulator control and was a hydraulic drive robot.
1961	Unimate robot installed at Ford Motor Company for tending a die casting machine.
1966	Trallfa, a Norwegian firm, built and installed a spray painting robot.
1968	A mobile robot named "Shakey" developed at SRI (Stanford Research Institute). It was equipped with a variety of sensors, including a vision camera and touch sensors, and can move on the floor.
1971	The "Stanford Arm," a small electrically powered robot arm, developed at Stanford University.
1973	First computer-type robot programming language developed at SRI for research called WAVE. Followed by the language AL in 1974. The two languages were subsequently developed into the commercial VAL, language for Unimation by Victor Scheinman and Bruce Simano.
1974	ASEA introduced the all-electric drive IRb6 robot.
1974	Kawasaki, under Unimation license, installed arc-welding operation for motorcycle frames.
1974	Cincinnati Milacron introduced the T3 robot with computer control.
1975	Olivetti "Sigma" robot used in assembly operation—one of the very first assembly applications of robotics.
1976	Remote Center Compliance (RCC) device for part insertion in assembly developed at Charles Stark Draper Labs in United States.

1978	PUMA (Programmable Universal Machine for Assembly) robot introduced for assembly by Unimation, based on designs from a General Motors study.
1978	Cincinnati Milacron T3 robot adapted and programmed to perform drilling and routing operations on aircraft components, under Air Force ICAM (Integrated Computer-Aided Manufacturing) sponsorship.
1979	Development of SCARA type robot (Selective Compliance Arm for Robotic Assembly) at Yamanashi University in Japan for assembly. Several commercial SCARA robots introduced around 1981.
1980	Bin-picking robotic system demonstrated at University of Rhode Island. Using machine vision, the system was capable of picking parts in random orientations and positions out of a bin.
1981	A "direct-drive robot" developed at Carnegie-Mellon University. It used electric motors located at the manipulator joints without the usual mechanical transmission linkages used on most robots.

Brief history of robotics

1982	IBM introduces the RS-1 robot for assembly, based on several years of in-house development. It is a box-frame robot, using an arm consisting of three orthogonal slides. The robot language AML, developed by IBM, also introduced to program the RS-1.
1983	Report issued on research at Westinghouse Corp under National Science Foundation sponsorship on "adaptable-programmable assembly system" (APAS), a pilot project for a flexible automated assembly line using wools.
1984	Several off-line programming systems demonstrated at the Robots 8 show. Typical operation of these systems allowed the robot program to be developed using interactive graphics on a personal computer and then downloaded to the robot.
1990's	Robot development diversified into walking robots at MIT, Honda, etc., rehabilitation robots for health care, as well as robots for defense and space applications.
2000's	Micro and nano robots using smart materials, Unmanned Aerial vehicles and underwater robotics.

Three generations of robotics / engineering

- **First generation of robots:** simple pick and place devices with no external sensors.
- **Second generation robots:** external Sensors (vision, tactile, etc) for interaction with the environment.
- **Third generation robots:** intelligence, smart materials, bio, etc.
- **Future robots:** bio-robots, micro, nano, cyborgs (cybernetic+organism), etc.

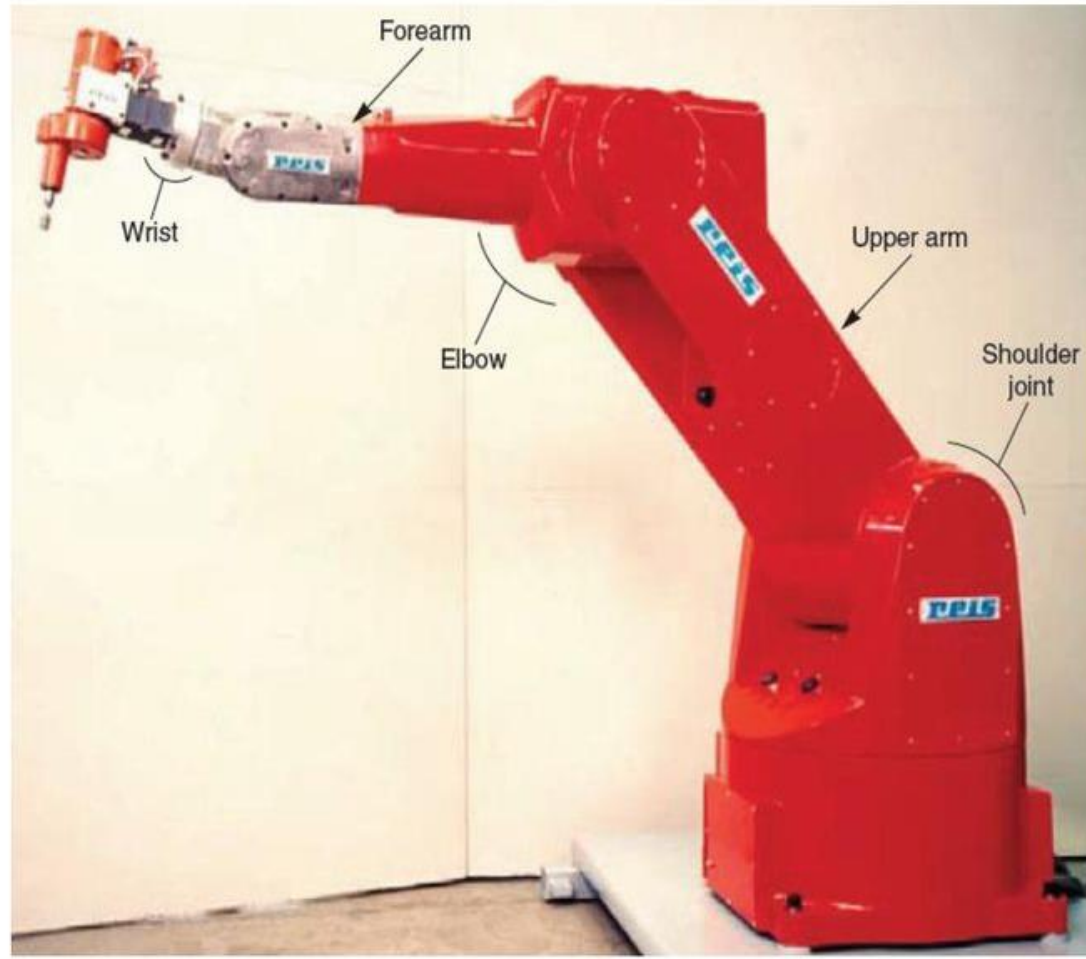
Basic components or elements of robot

A Robot is a system, consists of the following elements, which are integrated to form a whole:

- Manipulator / Rover : This is the main body of the Robot and consists of links, joints and structural elements of the Robot.
- Actuators : Actuators are the muscles of the manipulators. Common types of actuators are servomotors, stepper motors, pneumatic cylinders etc.
- Sensors : Sensors are used to collect information about the internal state of the robot or to communicate with the outside environment. Robots are often equipped with external sensory devices such as a vision system, touch and tactile sensors etc which help to communicate with the environment
- Controller : The controller receives data from the computer, controls the motions of the actuator and coordinates these motions with the sensory feedback information
- Software and Hardware: A computer software and hardware process through computation of the symbolic codes to derive the needed purpose of position, speed and accuracy of motion.

BASIC COMPONENTS OF ROBOT

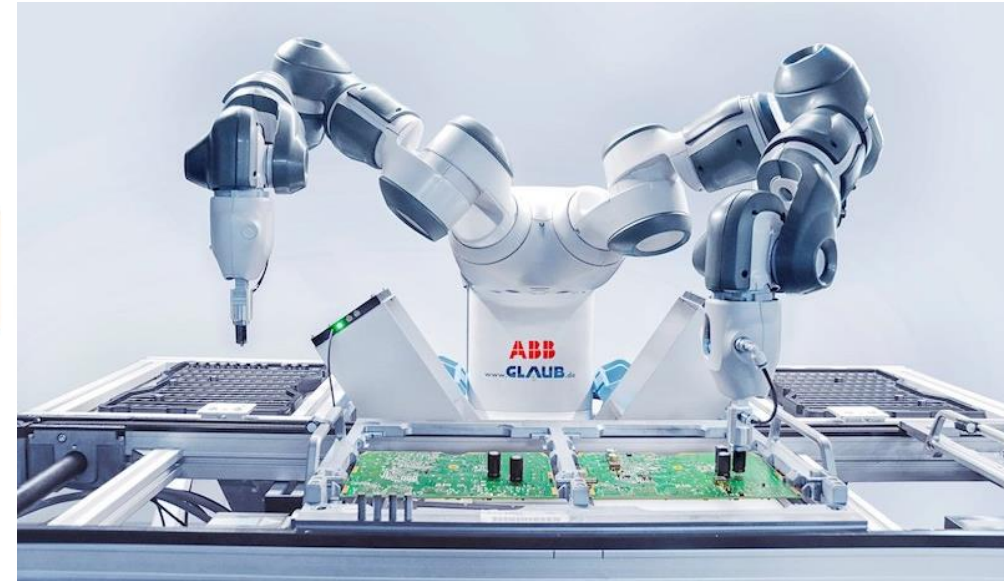
1. Arm
2. End Effectors
3. Actuators
4. Sensors
5. Controllers
6. Drive
7. Software



Robot specifications

Payload:

- The rated load carrying capacity of an industrial robot is defined by its weight of the object or the tool held by the gripper.
- These weight does not affect other functional characteristics like tip deflection, control of motion along defined path
- The overload may lead to the malfunctioning of the robot systems.



Robot specifications

Number of axes: Normally this is the number of DOF for a typical industrial serial robot.

Ex: SCARA Robot-4 Axis, FANUC R-2000iC- 6 Axis, KUKA iiwa- 7axis

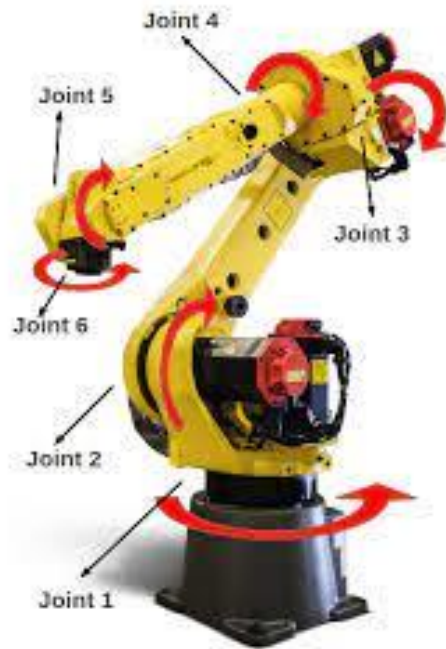
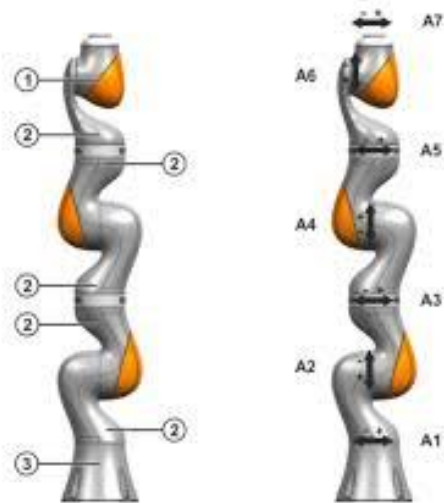


Table 2.2. General Robotic Axes

Type of axis	Axes	Arbitration
Major	1 to 3	Positioning the wrist
Minor	4 to 6	Orienting the gripper
Redundant	7 to i	Preventing obstacles



Speed: It is the distance moved by the tool tip in unit time.

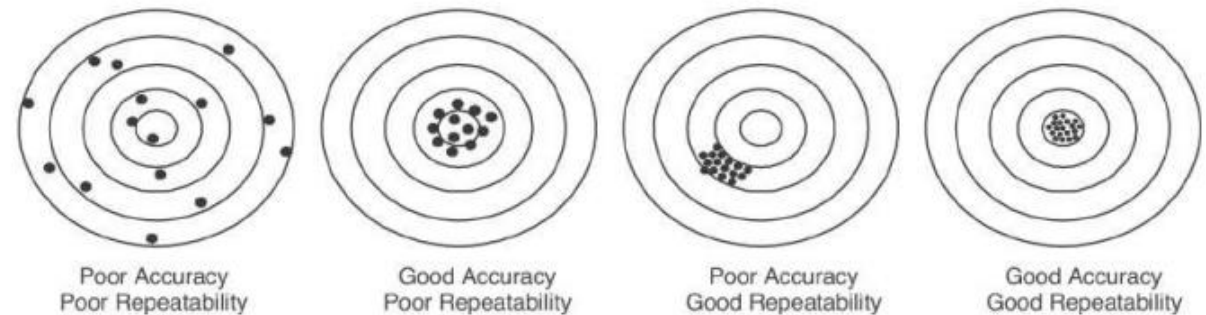
Robot specifications

Reach:

- The lengths of the links and configuration defines the reach of the industrial robot.
- The maximum and minimum extents of the robot positions give an idea about the reach of the robot
- It is also useful in the specification of the work envelope of the robot.

Precision of Robot: The reach of position of a robot is defined by resolution of actuators , sensors and control system this mainly depends on feed back systems.(Spatial resolution, Accuracy, Repeatability)

Accuracy: The accuracy of a manipulator is a measure of how close the manipulator can come to a given point within its workspace.



Robot specifications

Repeatability:

- The accuracy with which particular defined position can be *repeatedly* achieved by a robot is called *Repeatability*.
- Here positions have to be recorded and analyzed.
- The error which is obtained is later adjusted through programming.
- The primary method of sensing positioning errors in most cases is with position encoders located at the joints, either on the shaft of the motor that actuates the joint or on the joint itself. There is typically no direct measurement of the end-effector position and orientation.
- The actual programmed point will be different(+ or -) from target point due to limitation of control of resolution.
- The repeatability is affected by the condition of robot's components also

Robot specifications

Resolution:

- Also known as spatial resolution is the least count of the movement into which the robots work envelop can be divided to represent the incremental or decremental steps.
- It has two components Controller resolution and mechanical resolution.
- **Controller resolution means the smallest increment of motion that the controller can sense.**
- **Number of increments= 2^n**
- n = nob. of bits in control memory
- **For an example**, a robots with 8 bits of storage can divide the range into 256 discrete positions. The control resolution would be defined as total motions divided by number of increments (i.e., (range of motion)/256)
- The control resolution is determined by robot's position control system & feedback measurement system.
- It is the controllers ability to divide the total range of movements for particular joint into individual increments, that can be addressed in controller.
- The increments is also referred as addressable points

Robot specifications

Compliance:

- Compliance in robotics is the relationship between the forces and motion of a robot and the part it's working on.
- The term compliance refers to the displacement of the wrist end in response to force or torque exerted against it.
- It reduces robot's precision of the movement under load, if the bot is handling heavy load which may cause robot arm to deflect and also performance will be degraded.
- The term spring is sometime used to describe a robot with high compliance.

Types of Robots

Manipulators:

- Manipulator robots are typically robotic arms with multiple joints that can manipulate objects with precision.
- Commonly used in manufacturing, assembly lines, and research laboratories



Types of Robots

Legged robot:

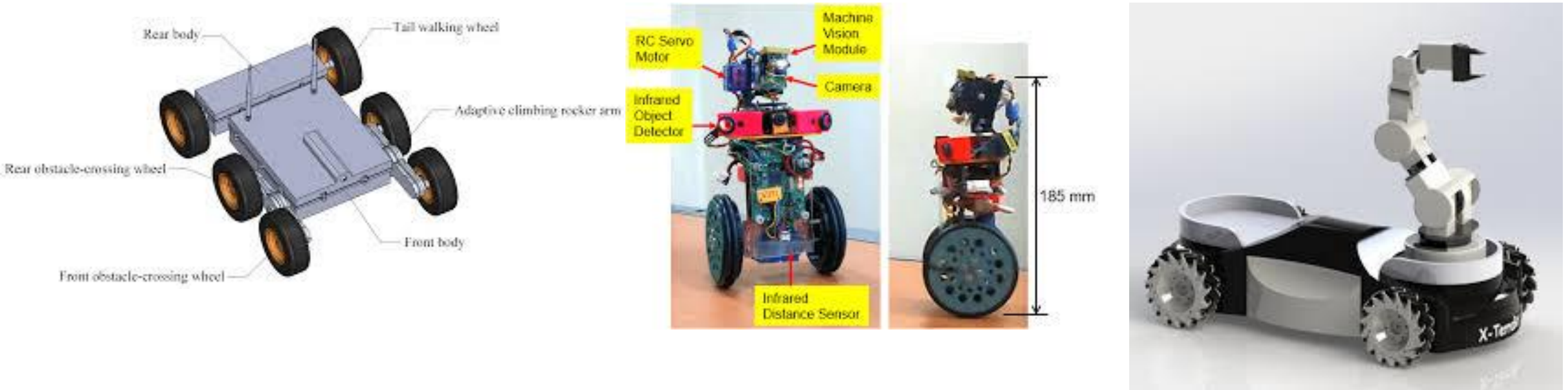
- Have legs or a combination of legs and wheels for locomotion.
- Designed to **navigate complex terrains** where wheeled robots may struggle, making them suitable for **search and rescue missions**, **exploration**, and **military applications**.



Types of Robots

Wheeled Robot:

- Wheeled robots are equipped with wheels for locomotion.
- Commonly used in applications such as logistics, surveillance, and planetary exploration, where smooth and efficient movement on flat surfaces is required



Types of Robots

Aerial Robots:

- Aerial robots, also known as **drones** or **unmanned aerial vehicles (UAVs)**
- are flying robots that can **navigate** through the air **autonomously** or **under remote control**.
- Utilized in diverse applications, including **aerial photography**, **surveillance**, **agriculture**, and **delivery services**.



Types of Robots

Humanoids:

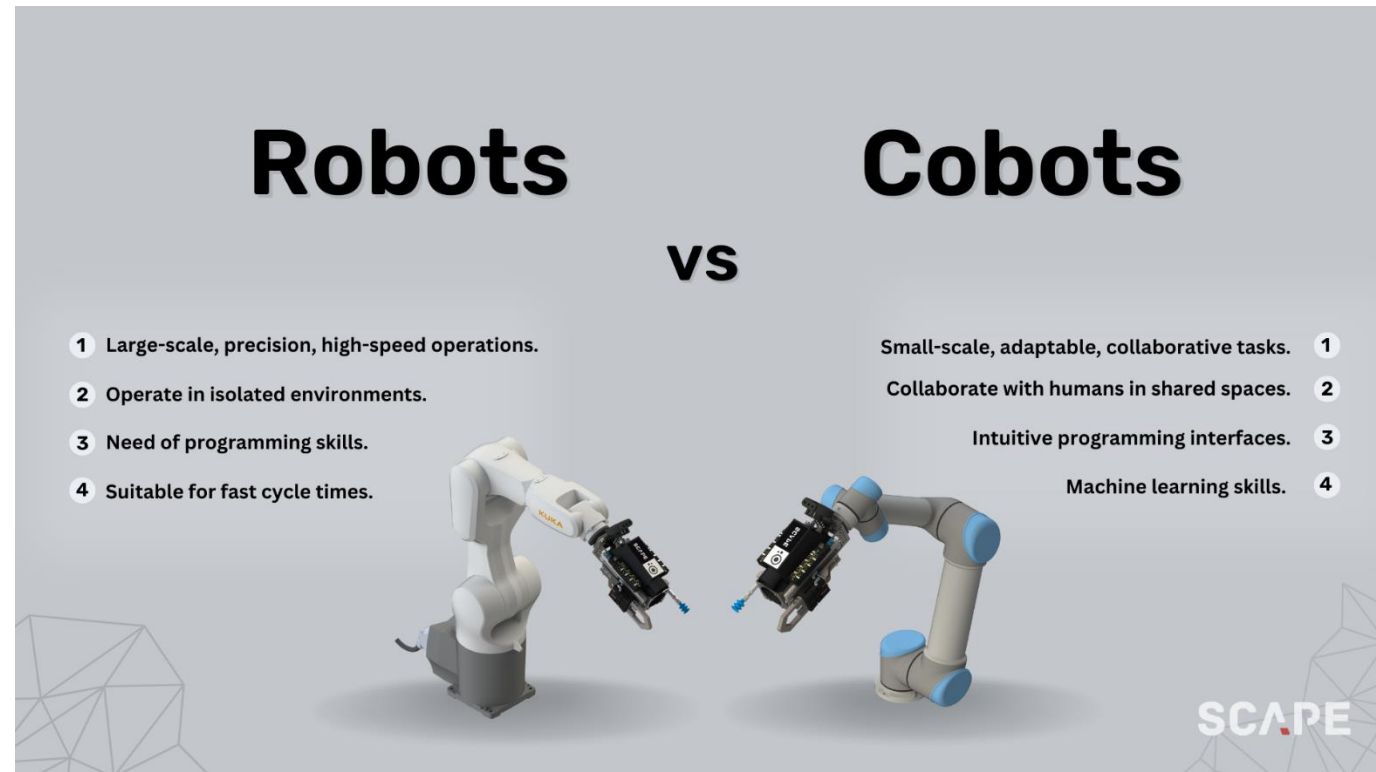
- Humanoid robots are designed to resemble and mimic human movements and behaviours to varying degrees.
- Used in research, entertainment, and assistive roles such as elderly care and rehabilitation.



Types of Robots

Cobots (Collaborative Robots):

- Designed to **work alongside humans** in shared workspaces, collaborating on tasks without safety barriers.
- Equipped with **sensors and advanced programming** to ensure **safe interaction** with humans.



Types of Robots

Autonomous Robots:

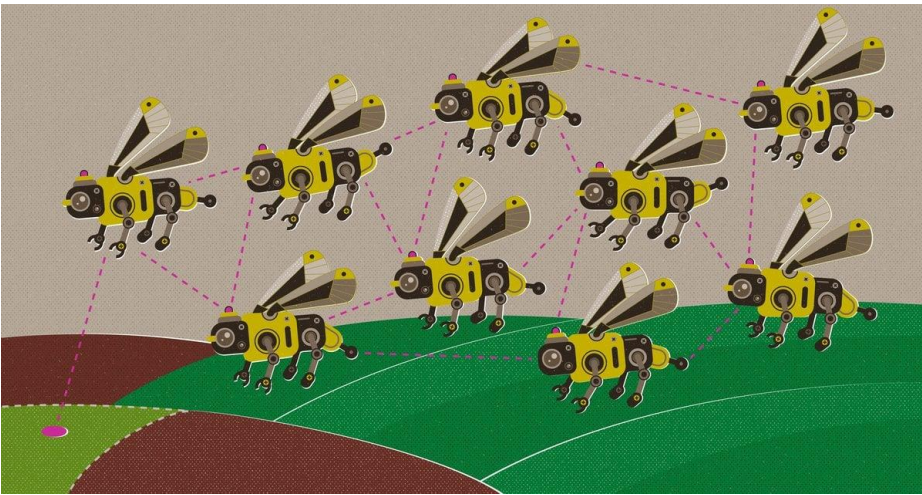
- Operate independently without direct human intervention, relying on sensors, algorithms, and decision-making capabilities to navigate and accomplish tasks.
- Used in various domains such as transportation, exploration, and environmental monitoring.



Types of Robots

Swarm Robots:

- A group of relatively simple robots that work together in a coordinated manner to accomplish tasks.
- They are inspired by collective behaviours observed in nature, such as swarms of insects or flocks of birds
- Used in applications like search and rescue, surveillance, and environmental monitoring.



Safety measures in robotics

- Before use:** Check for obstructions in the work area, make sure the robot is not damaged, and remove loose clothing and long hair. You should also know where the emergency stop button is and keep people out of the robot's workspace.
- Preventive measures:** Integrate preventive measures into the workspace, such as safety guards and emergency stop buttons to stop the robot in case of a hazard. You can also use physical barriers like fences and light curtains to define the robot's operational zone and prevent unauthorized access. Safety sensors and machine vision systems can detect humans nearby and trigger automatic shutdowns if needed.
- Passive safety systems:** Use passive safety systems like interlocks that prevent robots from operating when an operator enters the workspace. You can also use physical guards to protect operators from moving parts or machinery.
- Functional safety:** Implement automatic protection systems that can detect and mitigate malfunctions in the robot to keep humans safe.
- Operator training:** Provide comprehensive training for operators to reduce risks associated with hazards.

Social impact

- **Impact of Robotics on Direct labour:** Direct worker job loss, shift of direct labour to indirect labour, reducing the skill, training of workers to work on robots, mean time job opportunities open up in robot manufacturing industries.
- **Professional Adjustment Impact:** Employees adjusted to old techniques like process plans, personnel management, quality control, material movements etc., with advent of automation they have to familiar with programming, robot maintenance, optimization of processes, etc.,
- **Need for Education and training**

Robotics market and the future prospects

- **Robotics Market Insights:** Global Robotics Market size was valued at USD 34.06 billion in 2022 poised to grow from USD 39.54 Billion in 2023 to USD 134.64 Billion by 2031, growing at a Compound annual growth rate of 16.60% during the forecast period (2024-2031).
- Traditionally, it was the automotive industries that led the drive to robotize, however, in the last decade, the growth of non-automotive robots have been higher than automotive robots.
- One of the reason is fall in robotics prices as compared to increase in labor costs. At the same time, today's robots have better performance in terms of accuracy, speed, versatility and above all, computer power as compared to earlier robots.
- Other factors that would force industries to robotize is the shrinking of the work force in several developed countries and an ageing population.
- Robotics is a technology that can be harnessed solely for the benefit of humankind.
- However like other technologies, there are potential dangers involved and safeguards must be instituted to prevent its harmful use.

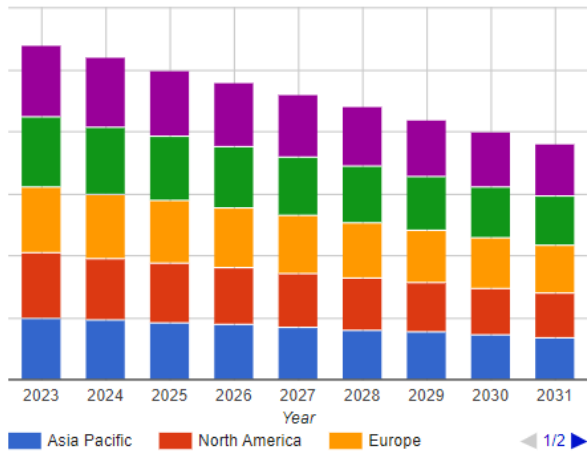
Robotics market and the future prospects

Market Snapshot - 2024-2031



Global Robotics Market (\$ Bn)

Country Share By North America (%)



Global Robotics Market By Robotics Type (%)

Articulated Robots Autonomous Mobile Robots Automated Guided Vehicle Humanoids Cobots Others



Advantages of robotics

- Environmental Safety
- Increased Productivity
- Reduced unit cost in long run and batches
- Replacement of robots in populated places
- Accuracy, repeatability and work quality
- Advanced accessories – Sensors, camera

Disadvantages of robotics

- Cost constraint in investment
- Decision Intelligence is complex
- Replacement of labour
- Real time responses have been the limitations on the programming capabilities.
- Degeneration of human skill and capability